

Appendix B. Stability analysis of equilibrium points

This appendix details the stability analysis supporting the mechanism claims in Section 4. In information governance terms, stability determines whether a governance configuration is self-reinforcing (lock-in), self-correcting (convergence), or unstable (oscillatory).

Appendix B.1. The Jacobian matrix

Stability is determined by the eigenvalues of the Jacobian matrix J of the system (Equation 7).

$$J = \begin{bmatrix} \frac{\partial \dot{\alpha}}{\partial \alpha} & \frac{\partial \dot{\alpha}}{\partial \beta} & \frac{\partial \dot{\alpha}}{\partial \gamma} \\ \frac{\partial \dot{\beta}}{\partial \alpha} & \frac{\partial \dot{\beta}}{\partial \beta} & \frac{\partial \dot{\beta}}{\partial \gamma} \\ \frac{\partial \dot{\gamma}}{\partial \alpha} & \frac{\partial \dot{\gamma}}{\partial \beta} & \frac{\partial \dot{\gamma}}{\partial \gamma} \end{bmatrix} \quad (\text{B.1})$$

For convenience, we define compact governance-gap terms corresponding to our mechanism interpretation: $\Delta_P = f_p + \gamma(1 - \beta)R_{p2} - C_{p2}$ (net accountability pressure), $\Delta_B = E_m + C_b + \gamma R_b + [\gamma + \alpha u_2(1 - \gamma)]f_{b1} + \alpha u_2 f_{b2} - (1 - u_1)(1 - \alpha u_2)E_u$ (net compliance incentive), and $\Delta_U = (1 - u_2\alpha)(1 - \beta)f_u - C_u$ (net reporting incentive). The Jacobian is as follows.

$$J = \begin{bmatrix} (1 - 2\alpha)\Delta_P & -\alpha(1 - \alpha)\gamma R_{p2} & \alpha(1 - \alpha)(1 - \beta)R_{p2} \\ \beta(1 - \beta)[u_2(1 - \gamma)f_{b1} + u_2 f_{b2} + u_2(1 - u_1)E_u] & (1 - 2\beta)\Delta_B & \beta(1 - \beta)[R_b + (1 - \alpha u_2)f_{b1}] \\ \gamma(1 - \gamma)[-u_2(1 - \beta)f_u] & \gamma(1 - \gamma)[-(1 - u_2\alpha)f_u] & (1 - 2\gamma)\Delta_U \end{bmatrix} \quad (\text{B.2})$$

Appendix B.2. Proofs for propositions 1–3

Proof of Proposition 1. Proposition 1 formalizes the dependence of downstream participation on incentive design (f_u, C_u) and upstream verification capacity (u_2). Stability depends on the sign of Δ_U . If $\Delta_U > 0$, $\gamma = 1$ is stable; if $\Delta_U < 0$, $\gamma = 0$ is stable. The threshold condition $\Delta_U = 0$ yields $\alpha_0 = \frac{1}{u_2}(1 - \frac{C_u}{(1 - \beta)f_u})$.

Proof of Proposition 2. This identifies the institutional baseline for accountability internalization: proactive governance becomes stable when passivity costs exceed verification costs. Stability depends on Δ_P . If $\Delta_P > 0$, $\alpha = 1$ is stable. The threshold is $\beta_0 = 1 - \frac{C_{p2} - f_p}{\gamma R_{p2}}$.

Proof of Proposition 3. This operationalizes the technological core of deterrence: verification and sanctions determine the economic viability of illicit creation. Stability depends on Δ_B . If $\Delta_B > 0$, compliant creation ($\beta = 1$) is stable.

Appendix B.3. Eigenvalue analysis for each equilibrium point

Eigenvalues ($\lambda_1, \lambda_2, \lambda_3$) are the diagonal elements of the Jacobian evaluated at equilibrium points.

Example: Stability of $E_2(0, 0, 1)$. Substituting $\alpha = 0, \beta = 0, \gamma = 1$ yields:

- $\lambda_1 = f_p + R_{p2} - C_{p2}$
- $\lambda_2 = E_m + C_b + R_b + f_{b1} - (1 - u_1)E_u$
- $\lambda_3 = C_u - f_u$

This links directly to the user-led content governance trap: once $\gamma = 1$ stabilizes ($f_u > C_u \Rightarrow \lambda_3 < 0$), system stability hinges on whether incentives shift platform and creator behaviors ($\lambda_1 < 0, \lambda_2 < 0$).